

2.2 Related Rates

When two or more quantities that change with time are linked by an equation, that equation can be differentiated with respect to time to produce an equation linking the rates of change of the quantities. Below we give the procedure for dealing with such problems

1. From the given data make a sketch if possible.
2. Define symbols you want to use that are not defined in the statement of the problem. Express given and required quantities and rates in terms of these symbols.
3. Identify one or more equations linking the variable quantities.
4. Differentiate the equation(s) implicitly with respect to time, regarding variable quantities as functions of time.
5. Substitute any given values for the quantities and their rates, then solve the resulting equation(s) for the unknown quantities and rates.
6. Make a concluding statement answering the question asked.

■ **Example 2.2.1** An aircraft is flying horizontally at a speed of 600 km/h . How fast is the distance between the aircraft and a radio beacon increasing 1 minute after the aircraft passes 5 km directly above the beacon? ■

Solution

■ **Example 2.2.2** How fast is the area of a rectangle changing if one side is 10 cm long and is increasing at a rate of 2 cm/s and the other side is 8 cm long and is decreasing at a rate of 3 cm/s ? ■

Solution

Exercise 2.2

1. Find the rate of change of the area of a square whose side is 8 cm long, if the side length is increasing at 2 cm/min .
2. The area of a square is decreasing at $2\text{ ft}^2/\text{s}$. How fast is the side length changing when it is 8 ft ?
3. A pebble dropped into a pond causes a circular ripple to expand outward from the point of impact. How fast is the area enclosed by the ripple increasing when the radius is 20 cm and is increasing at a rate of 4 cm/s .
4. Air is being pumped into a spherical balloon. The volume of the balloon is increasing at a rate of $20\text{ cm}^3/\text{s}$ when the radius is 30 cm . How fast is the radius increasing at that time?
5. The volume of a right circular cylinder is 60 cm^3 and is increasing at $2\text{ cm}^3/\text{min}$ at a time when the radius is 5 cm and is increasing at 1 cm/min . How fast is the height of the cylinder changing at that time?
6. A point moves on the curve $y = x^2$. How fast is y changing when $x = -2$ and x is decreasing at a rate of 3?

2.3 Extreme Values

Intervals of Increase and Decrease

In the discussion of extreme values of a function and indeed in a number of concepts of applications of differentiation it is necessary to use the concepts of intervals of increase and decrease of functions. We begin by giving the respective definitions and theorems that give the strategies of determining intervals of increase and decrease of functions.

Definition 2.3.1 — Increasing and decreasing functions. Suppose that the function f is defined on an interval I and that x_1 and x_2 are two points in I .

- (a) If $f(x_2) > f(x_1)$ whenever $x_2 > x_1$, we say that f is increasing on I .
- (b) If $f(x_2) < f(x_1)$ whenever $x_2 > x_1$, we say that f is decreasing on I .
- (c) If $f(x_2) \geq f(x_1)$ whenever $x_2 > x_1$, we say that f is non-decreasing on I .
- (d) If $f(x_2) \leq f(x_1)$ whenever $x_2 > x_1$, we say that f is non-increasing on I .

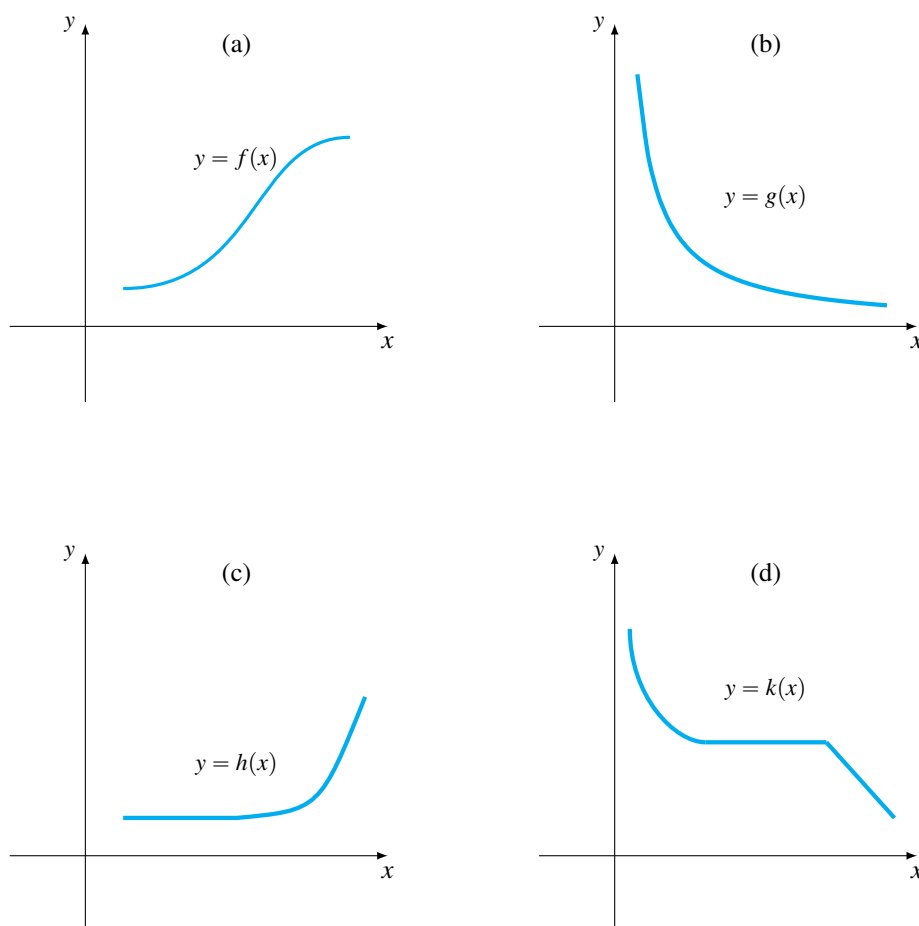


Figure 2.2: (a) Function f is increasing (b) Function g is decreasing (c) Function h is nondecreasing (d) Function k is nonincreasing

In the following theorem we state the criteria of determining whether a function is increasing or decreasing on a given interval using the first derivative. This proves to be handy than using the definition in every circumstance.

Theorem 2.3.1 Let J be an open interval and let I be an interval consisting of all points in J and possibly one or both of the end points of J . Suppose that f is continuous on I and differentiable on J .

- (a) If $f'(x) > 0$ for all $x \in J$, then f is increasing on I .
- (b) If $f'(x) < 0$ for all $x \in J$, then f is decreasing on I .
- (c) If $f'(x) \geq 0$ for all $x \in J$, then f is non-decreasing on I .
- (d) If $f'(x) \leq 0$ for all $x \in J$, then f is non-increasing on I .

■ **Example 2.3.1** On what intervals is the function $f(x) = x^3 - 12x + 1$ increasing? On what intervals is it decreasing? ■

Solution

■ **Example 2.3.2** Show that $f(x) = x^3$ is increasing on any interval. ■

Solution

Solution**Maximum and Minimum Values**

Extreme values whenever they exist are in two categories, global (with respect to the domain of a given function) or local (with respect to a given point in the domain). We start by defining the global extreme values also called absolute extreme values.

Definition 2.3.2 — Absolute extreme values. Function f has an **absolute maximum value** $f(x_0)$ at x_0 in its domain if $f(x) \leq f(x_0)$ holds for every x in the domain of f .

Similarly, f has an **absolute minimum value** $f(x_0)$ at x_0 in its domain if $f(x) \geq f(x_0)$ holds for every x in the domain of f .



1. A function will have only one absolute maximum (or minimum) value if it exists. However the value can occur at many points. For example, $f(x) = \sin x$ has absolute maximum of 1 but it occurs at every point $\frac{\pi}{2} + 2n\pi, n \in \mathbb{Z}$.
2. A function need not have any extreme value. The function $f(x) = \frac{1}{x}$ becomes arbitrarily large as x approaches 0 from the right, and so has no finite absolute maximum value.

When is it guaranteed that an absolute extreme value does exist? We see in the following theorem that this a functionality of both the function and its domain.

Theorem 2.3.2 If the domain of the function f is a **closed, finite interval** or a union of finitely many such intervals, and if f is continuous on that domain, then f must have an absolute maximum value and absolute minimum value.

Definition 2.3.3 — Local extreme values. Function f has a **local maximum value** (loc. max.) $f(x_0)$ at the point x_0 in its domain provided $\exists$ a number $h > 0$ such that $f(x) \leq f(x_0)$ whenever x is in the domain of f and $|x - x_0| < h$.

Similarly, f has a **local minimum value** (loc. min.) $f(x_1)$ at the point x_1 in its domain provided $\exists$ a number $h > 0$ such that $f(x) \geq f(x_1)$ whenever x is in the domain of f and $|x - x_1| < h$.

What points do qualify for investigation for the existence of points where extreme values are attained. To discuss this lets look at Figure 2.3.

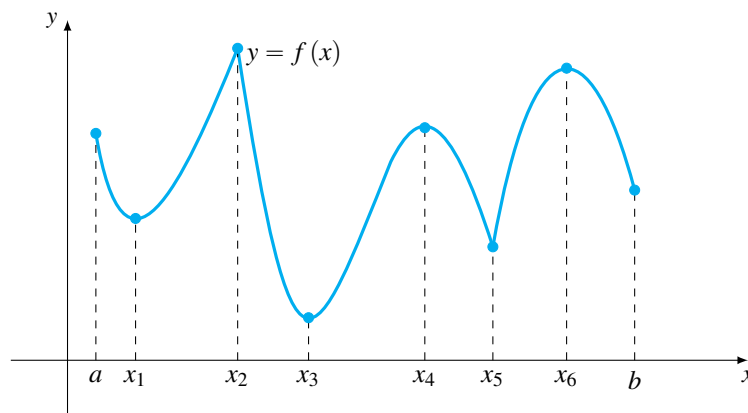


Figure 2.3: Local extreme values

From the figure we see that local extreme values can occur at any of the following points.

- (i) **stationary points** of f ; points $x \in D(f)$ where $f'(x) = 0$.
- (ii) **singular points** of f ; points $x \in D(f)$ where $f'(x)$ is not defined.
- (iii) **endpoints** of the domain of f ; points that do belong to $D(f)$ but are not interior points of $D(f)$.

In the figure above, x_1, x_3, x_4 and x_6 are stationary points, x_2 and x_5 are singular points and a and b are endpoints.

Theorem 2.3.3 If the function f is defined on an interval I and has a local maximum (or local minimum) value at the point $x = x_0$ in I , then x_0 must be either a critical point of f , a singular point of f , or an endpoint of I .

■ **Example 2.3.3** Find the maximum and minimum values of the function $g(x) = x^3 - 3x^2 - 9x + 2$ on the interval $-2 \leq x \leq 2$. ■

Solution

■ **Example 2.3.4** Find and classify the local and absolute extreme values of the function $f(x) = x - x^{2/3}$ on the interval $[-1, 2]$. Sketch the graph. ■

Solution

 If f' is positive (or negative) on both sides of a stationary or singular point, then f has neither a maximum nor a minimum value at that point.

■ **Example 2.3.5** Find the local and absolute extreme values of $f(x) = x^4 - 2x^2 - 3$ on the interval $[-2, 2]$. Sketch the graph of f . ■

Solution

In the next theorem we state the strategy used to determine the existence of extreme values in the case where the domain is an open interval.

Theorem 2.3.4 — Existence of extreme values on open intervals. If f is continuous on an open interval (a, b) , and if

$$\lim_{x \rightarrow a^+} f(x) = L \text{ and } \lim_{x \rightarrow b^-} f(x) = M$$

then the following conclusions hold

- (i) If $f(u) > L$ and $f(u) > M$ for some $u \in (a, b)$, then f has an absolute maximum value on (a, b) .
- (ii) If $f(u) < L$ and $f(u) < M$ for some $u \in (a, b)$, then f has an absolute minimum value on (a, b) .

In this theorem a may be $-\infty$ and b may ∞ . Also, either or both L and M may be ∞ or $-\infty$.

■ **Example 2.3.6** Show that $f(x) = x + \frac{4}{x}$ has an absolute minimum value on the interval $(0, \infty)$, and find that minimum value. ■

Solution